

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA14

COURSE NAME: Compiler Design

COURSE OUTCOME COVERED

CO4: Examine intermediate code generation techniques to enhance code optimization (BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 4

Total Marks:40

Annexure A

CLASS TEST

Code of Conduct:

I M.VINAY(192311146) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

M.VINAY

Signature of the student:

Q. No	Understanding of Concepts (25)	Explanation (25)	Application (20)	Organization(15)	Accuracy(10)	Total Score (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
OVERALL RUBRICS VALUE						
	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	/ 30	/ 20	/ 15	/ 10	/ 100

Annexure A

CLASS TEST

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	15%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps

1) $((a < b) \& (c < d) \text{ or } (a > d)))$

$z = x + y * z$

else

$z = z + 1$

Steps:-

L1:-

if $a < b$ goto L2
goto L5

L2:

if $c < d$ goto L4
goto L3

L3:

if $a > d$ goto L4
goto L5

L4:

$T1 = y * z$

$T2 = x + T1$

$z = T2$

goto L6

L5:

$T3 = z + 1$

$z = T3$

L6:

2) $(a < b) \& (c < d) \text{ or } (e < f)$

1. if $a < b$ goto 3

2. goto 5

3. if $c < d$ goto 7

4. goto 5

5. if $e < f$ goto 7

6. goto 8

7. TRUE

8. FALSE

3)

$$A = (-A + B) * (C - A + B) * C$$

$$t1 = -A$$

$$t2 = t1 + B$$

$$t3 = -A$$

$$t4 = t3 + B$$

$$t5 = t4 * C$$

$$t6 = t2 * t5$$

$$A = t6$$

S.No	operator	Arg1	Arg 2	result
0	unimus	A	-	t ₁
1	+	t ₁	B	t ₂
2	unimus	A	-	t ₃
3	+	t ₃	B	t ₄
4	*	t ₄	C	t ₅
5	*	t ₂	t ₅	t ₆
6	=	t ₆	-	A

4) $(a > b) \text{ or } (c < d) \text{ or } (e < f)$

L1:

if $a > b$ goto L TRUE
goto L2

L2:

if $c < d$ goto L TRUE
goto L3

L3:

if $e < f$ goto L TRUE
goto FALSE

L TRUE

boolean expression is TRUE
goto END

L FALSE

boolean expression is FALSE
LEND